CHAPTER 12 · DERIVATIONS

# ELECTROSTATICS

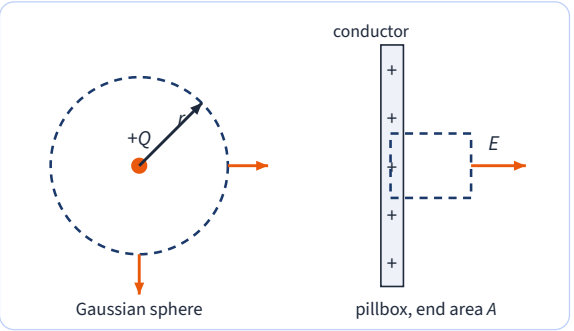
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derivation the STPM 2026 syllabus asks you to derive, deduce or show

## 1 ELECTRIC FIELD STRENGTH FROM GAUSS'S LAW

SYLLABUS 13.3(D)

**RESULT** $E = \dfrac{Q}{4\pi\varepsilon_0 r^2}$; sphere: 0 inside; plate: $E = \dfrac{\sigma}{\varepsilon_0}$

1. **Point charge Q**: Gaussian sphere of radius r centred on Q. E is radial and the same everywhere on it, so $\Phi = E(4\pi r^2) = \dfrac{Q}{\varepsilon_0}$, giving $\mathbf{E = \dfrac{Q}{4\pi\varepsilon_0 r^2}}$.
2. **Charged conducting sphere** (radius R, charge Q on its surface): for r > R the Gaussian sphere encloses Q, so $E = \dfrac{Q}{4\pi\varepsilon_0 r^2}$; for r < R it encloses no charge, so **E = 0 inside**.
3. **Charged conducting plate** (surface charge density σ): a pillbox of end area A crosses the surface, one end inside the conductor (E = 0) and one end outside.
4. Flux only through the outer end: $EA = \dfrac{\sigma A}{\varepsilon_0}$, so $\mathbf{E = \dfrac{\sigma}{\varepsilon_0}}$, perpendicular to the surface.



CHAPTER 13 · DERIVATIONS

# CAPACITORS

4 derivations the STPM 2026 syllabus asks you to derive, deduce or show

## 1 CAPACITANCE OF A PARALLEL PLATE CAPACITOR

SYLLABUS 14.1(D)

**RESULT** $C_0 = \dfrac{\varepsilon_0 A}{d}$

1. Plates of area A, separation d, charges ±Q: $\sigma = \dfrac{Q}{A}$.
2. Gauss's law: the field between the plates is $E = \dfrac{\sigma}{\varepsilon_0} = \dfrac{Q}{\varepsilon_0 A}$ (uniform).
3. $V = Ed = Qd/(\varepsilon_0 A)$.
4. $\mathbf{C_0 = \dfrac{Q}{V} = \dfrac{\varepsilon_0 A}{d}}$.

## 2 CAPACITORS IN SERIES AND IN PARALLEL

SYLLABUS 14.3(A)

**RESULT** series: $\dfrac{1}{C} = \dfrac{1}{C_1} + \dfrac{1}{C_2} + \ldots$; parallel: $C = C_1 + C_2 + \ldots$

1. **Series**: each capacitor carries the same charge Q, and $V = V_1 + V_2 + V_3$.
2. $\dfrac{Q}{C} = \dfrac{Q}{C_1} + \dfrac{Q}{C_2} + \dfrac{Q}{C_3}$, so $\mathbf{\dfrac{1}{C} = \dfrac{1}{C_1} + \dfrac{1}{C_2} + \dfrac{1}{C_3}}$.
3. **Parallel**: each capacitor has the same p.d. V, and $Q = Q_1 + Q_2 + Q_3$.
4. $CV = C_1V + C_2V + C_3V$, so $\mathbf{C = C_1 + C_2 + C_3}$.

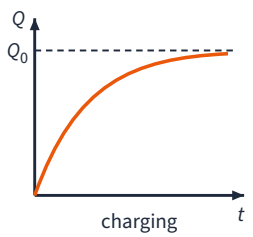
Energy formulae $U = \frac{1}{2}QV = \frac{1}{2}CV^2 = \dfrac{Q^2}{2C}$: derivation not required (14.4).

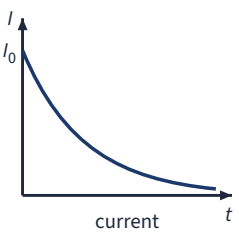
## 3 CHARGING THROUGH A RESISTOR

SYLLABUS 14.5(B)

**RESULT** $Q = Q_0(1 - e^{-t/RC})$, $V = V_0(1 - e^{-t/RC})$, $I = I_0 e^{-t/RC}$

1. Kirchhoff's law round the circuit (supply $V_0$): $V_0 = IR + \dfrac{Q}{C}$, with $I = \dfrac{dQ}{dt}$.
2. $R\dfrac{dQ}{dt} = \dfrac{CV_0 - Q}{C}$, so $\dfrac{dQ}{CV_0 - Q} = \dfrac{dt}{RC}$.
3. Integrate from Q = 0 at t = 0: $-\ln\dfrac{CV_0 - Q}{CV_0} = \dfrac{t}{RC}$.
4. $\mathbf{Q = Q_0(1 - e^{-t/RC})}$, with $Q_0 = CV_0$; dividing by C: $\mathbf{V = V_0(1 - e^{-t/RC})}$.
5. $I = \dfrac{dQ}{dt} = \dfrac{Q_0}{RC}e^{-t/RC}$: $\mathbf{I = I_0 e^{-t/RC}}$, with $I_0 = \dfrac{V_0}{R}$.



## 4 DISCHARGING THROUGH A RESISTOR

SYLLABUS 14.5(D)

**RESULT** $Q = Q_0e^{-t/RC}$, $V = V_0e^{-t/RC}$, $I = I_0e^{-t/RC}$

1. With the supply removed: $0 = IR + \frac{Q}{C}$, and $I = \frac{dQ}{dt}$.
2. $\frac{dQ}{dt} = -\frac{Q}{RC}$, so $\frac{dQ}{Q} = -\frac{dt}{RC}$.
3. Integrate from $Q_0$ at $t = 0$: $\ln\frac{Q}{Q_0} = -\frac{t}{RC}$, so $\mathbf{Q = Q_0e^{-t/RC}}$ and $\mathbf{V = V_0e^{-t/RC}}$.
4. Size of the current: $I = Q_0e^{-t/RC}/RC$, i.e. $\mathbf{I = I_0e^{-t/RC}}$ (in the opposite direction to the charging current).

CHAPTER 14 · DERIVATIONS

# ELECTRIC CURRENT

3 derivations the STPM 2026 syllabus asks you to derive, deduce or show

## 1 CURRENT AND DRIFT VELOCITY

SYLLABUS 15.2(B)

**RESULT** $I = Anev$

1. Wire of cross-sectional area A, n free electrons per unit volume, drift velocity v.
2. In time t the electrons move a distance vt, so all electrons in the volume Avt pass a cross-section.
3. Number = nAvt; charge Q = nAvte.
4. $\mathbf{I = \frac{Q}{t} = Anev}$.

## 2 OHM'S LAW AND J = ΣE

SYLLABUS 15.4(E)

**RESULT** $J = \sigma E \Leftrightarrow V = IR$

1. Uniform conductor of length l and area A with p.d. V: $J = \frac{I}{A}$ and $E = \frac{V}{l}$.
2. $J = \sigma E$ gives $\frac{I}{A} = \frac{\sigma V}{l}$.
3. $V = I\frac{l}{\sigma A} = I\frac{\rho l}{A} = IR$.
4. For a given conductor at constant temperature R is constant, so $V \propto I$: Ohm's law.

## 3 CONDUCTIVITY

SYLLABUS 15.4(F)

**RESULT** $\sigma = \frac{ne^2t}{m}$

1. In a field E an electron has acceleration $a = \frac{eE}{m}$.
2. Starting from zero average velocity after each collision, the mean time between collisions is t, so the drift velocity $v = at = \frac{eEt}{m}$.
3. $J = nev = \frac{ne^2t}{m}E$.
4. Comparing with $J = \sigma E$: $\mathbf{\sigma = \frac{ne^2t}{m}}$.

CHAPTER 16 · DERIVATIONS

# MAGNETIC FIELDS

5 derivations the STPM 2026 syllabus asks you to derive, deduce or show

## 1 FORCE ON A CURRENT-CARRYING CONDUCTOR

SYLLABUS 17.3(B)

**RESULT** $F = IlB \sin\theta$

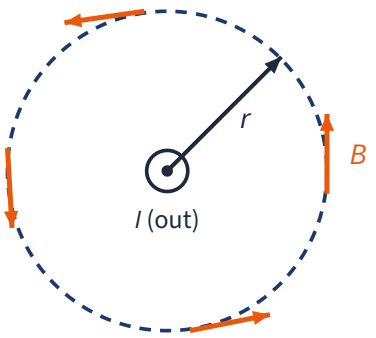
1. Each charge carrier q drifts with velocity v at angle θ to B: force $qvB \sin\theta$.
2. Number of carriers in length l: nAl.
3. Total force $F = (nAl)qvB \sin\theta$.
4. $I = nAqv$, so $\mathbf{F = IlB \sin\theta}$.

## 2 FIELD OF A LONG STRAIGHT WIRE (AMPÈRE'S LAW)

SYLLABUS 17.4(B)

**RESULT** $B = \dfrac{\mu_0 I}{2\pi r}$

1. Ampère's law: $\oint B \cdot dl = \mu_0 I$.
2. Choose a circle of radius r centred on the wire. By symmetry B is tangential and has the same size all round.
3. $\oint B \cdot dl = B(2\pi r)$.
4. $B(2\pi r) = \mu_0 I$, so $\mathbf{B = \dfrac{\mu_0 I}{2\pi r}}$.




## 3 FORCE BETWEEN TWO PARALLEL CONDUCTORS

SYLLABUS 17.5(A)

**RESULT** $F = \dfrac{\mu_0 I_1 I_2 l}{2\pi d}$

1. Field of wire 1 at wire 2 (distance d): $B_1 = \dfrac{\mu_0 I_1}{2\pi d}$, perpendicular to wire 2.
2. Force on length l of wire 2: $F = B_1 I_2 l$.
3. $\mathbf{F = \dfrac{\mu_0 I_1 I_2 l}{2\pi d}}$; an equal and opposite force acts on wire 1.
4. Currents in the same direction attract; opposite directions repel.

## 4 E/M IN THOMSON'S EXPERIMENT

SYLLABUS 17.6(C)

**RESULT** $\frac{e}{m} = \frac{E}{rB^2}$

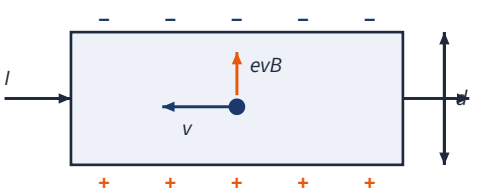
1. Crossed E and B fields are adjusted so the beam is undeflected: eE = evB, so $v = \frac{E}{B}$.
2. With E switched off, the electrons move in a circle of radius r in B: $evB = \frac{mv^2}{r}$.
3. $\frac{e}{m} = \frac{v}{rB}$.
4. Substitute $v = \frac{E}{B}$: $\mathbf{\frac{e}{m} = \frac{E}{rB^2}}$.

## 5 HALL VOLTAGE

SYLLABUS 17.7(B)

**RESULT** $V_H = \frac{BI}{net}$

1. Carriers (charge e, number density n) drift with velocity v through a slab of width d and thickness t, with B perpendicular.
2. The magnetic force evB pushes carriers to one face; charge builds up and sets up a Hall field $E_H = \frac{V_H}{d}$.
3. Equilibrium: $\frac{eV_H}{d} = evB$, so $V_H = Bvd$.
4. $I = neAv = ne(dt)v$, so $v = I/(nedt)$.
5. $\mathbf{V_H = \frac{BI}{net}}$.




*B* into page; $V_H$ across *d*; thickness *t*

CHAPTER 17 · DERIVATIONS

# ELECTROMAGNETIC INDUCTION

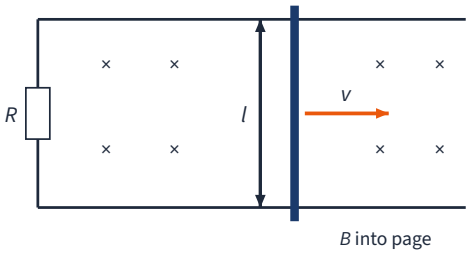
4 derivations the STPM 2026 syllabus asks you to derive, deduce or show

## 1 E.M.F. IN A MOVING STRAIGHT CONDUCTOR

SYLLABUS 18.2(C)

**RESULT** $\varepsilon = Blv \sin\theta$

1. A rod of length l moves with velocity v at angle θ to the field B (rod perpendicular to B).
2. In time dt it sweeps an area $l(v \sin\theta)dt$ across the field lines.
3. Flux cut $d\Phi = Blv \sin\theta\, dt$.
4. Faraday's law: $\boldsymbol{\varepsilon = \frac{d\Phi}{dt} = Blv \sin\theta}$.




## 2 E.M.F. IN A ROTATING COIL

SYLLABUS 18.2(D)

**RESULT** $\varepsilon = NBA\omega \sin\omega t$

1. A coil of N turns and area A rotates at angular velocity ω about an axis perpendicular to B.
2. Starting with its plane perpendicular to B, the angle between B and the normal is $\theta = \omega t$.
3. Flux linkage $N\Phi = NBA \cos\omega t$.
4. $\boldsymbol{\varepsilon = -\frac{d(N\Phi)}{dt} = NBA\omega \sin\omega t}$; peak value $\varepsilon_0 = NBA\omega$.

## 3 SELF-INDUCTANCE OF A SOLENOID

SYLLABUS 18.3(D)

**RESULT** $L = \frac{\mu_0 N^2 A}{l}$

1. Long solenoid: N turns, length l, area A, current I: $B = \frac{\mu_0 NI}{l}$.
2. Flux through each turn $\Phi = BA = \frac{\mu_0 NIA}{l}$.
3. Flux linkage $N\Phi = \frac{\mu_0 N^2 AI}{l}$.
4. $LI = N\Phi$, so $\boldsymbol{L = \frac{\mu_0 N^2 A}{l}}$.

## 4 MUTUAL INDUCTANCE OF TWO COAXIAL SOLENOIDS

SYLLABUS 18.5(D)

**RESULT** $M = \dfrac{\mu_0 N_p N_s A}{l_p}$

1. Primary solenoid ($N_p$ turns, length $l_p$) carries $I_p$: $B = \dfrac{\mu_0 N_p I_p}{l_p}$.
2. The secondary ($N_s$ turns) is wound coaxially with the same area A: $\Phi_s = BA$.
3. $N_s\Phi_s = \dfrac{\mu_0 N_p N_s A I_p}{l_p}$.
4. $MI_p = N_s\Phi_s$, so $\mathbf{M = \dfrac{\mu_0 N_p N_s A}{l_p}}$.

CHAPTER 18 · DERIVATIONS

# ALTERNATING CURRENT CIRCUITS

4 derivations the STPM 2026 syllabus asks you to derive, deduce or show

## 1 PURE RESISTOR: CURRENT AND POWER

SYLLABUS 19.1(C), (E)

**RESULT** $I = I_0 \sin \omega t$ (in phase); $\langle P \rangle = \frac{1}{2}V_0 I_0 = I_{rms}^2 R$

1. $V = V_0 \sin \omega t$ across R: $I = \frac{V}{R} = \frac{V_0}{R} \sin \omega t = \mathbf{I_0 \sin \omega t}$, in phase with V.
2. Instantaneous power $P = VI = V_0 I_0 \sin^2 \omega t = \frac{1}{2}V_0 I_0 (1 - \cos 2\omega t)$.
3. The mean of $\cos 2\omega t$ over a cycle is zero: $\mathbf{\langle P \rangle = \frac{1}{2}V_0 I_0 = I_{rms} V_{rms} = I_{rms}^2 R}$.

## 2 PURE INDUCTOR: CURRENT, REACTANCE AND POWER

SYLLABUS 19.2(A), (D), (E)

**RESULT** $I = I_0 \sin(\omega t - \frac{\pi}{2})$; $X_L = \omega L$; $\langle P \rangle = 0$

1. $V = V_0 \sin \omega t$ is balanced by the back e.m.f.: $V_0 \sin \omega t = L \frac{dI}{dt}$.
2. Integrate: $I = -\frac{V_0}{\omega L} \cos \omega t = \mathbf{I_0 \sin(\omega t - \frac{\pi}{2})}$: the current lags the voltage by $\frac{\pi}{2}$.
3. $X_L = \frac{V_0}{I_0} = \mathbf{\omega L}$.
4. $P = VI = -V_0 I_0 \sin \omega t \cos \omega t = -\frac{1}{2}V_0 I_0 \sin 2\omega t$; its mean over a cycle is **zero**.

## 3 PURE CAPACITOR: CURRENT, REACTANCE AND POWER

SYLLABUS 19.3(A), (D), (E)

**RESULT** $I = I_0 \sin(\omega t + \frac{\pi}{2})$; $X_C = \frac{1}{\omega C}$; $\langle P \rangle = 0$

1. $Q = CV = CV_0 \sin \omega t$.
2. $I = \frac{dQ}{dt} = \omega C V_0 \cos \omega t = \mathbf{I_0 \sin(\omega t + \frac{\pi}{2})}$: the current leads the voltage by $\frac{\pi}{2}$.
3. $X_C = \frac{V_0}{I_0} = \mathbf{\frac{1}{\omega C}}$.
4. $P = VI = \frac{1}{2}V_0 I_0 \sin 2\omega t$; its mean over a cycle is **zero**.

## 4 RESONANCE FREQUENCY OF AN R-L-C CIRCUIT

SYLLABUS 19.4(F)

**RESULT** $f_0 = 1/(2\pi\sqrt{LC})$

1. $Z = \sqrt{R^2 + (X_L - X_C)^2}$ is smallest ($Z = R$) when $X_L = X_C$.
2. $\omega_0 L = \dfrac{1}{\omega_0 C}$, so $\omega_0 = \dfrac{1}{\sqrt{LC}}$.
3. **$f_0 = 1/(2\pi\sqrt{LC})$**; the current is then maximum and in phase with the supply.